

Paper Audit Questions

Fluent, formatted output reads like expertise. These five questions check whether it earned the climb.

Run it on **any confident claim** — an AI answer, a study, the next vendor pitch, a student's brief, the AI-written case write-up someone hands you. Ten minutes, no new software.

THE FIVE QUESTIONS

1 Is the comparison fair?

The length trap, the apples-to-oranges baseline. *What is actually being held constant — and what isn't?*

2 Did they measure what they claimed?

Validated instrument? Any accuracy test? For a case: did it reason from *this* patient, or pattern-match to a type?

3 Who decided — and what's their stake?

Conflicted or independent judges? Whose voices produced this — and *whose were left out?*

4 What did the headline drop from the source?

Trace it back up the chain — claim → press release → paper → data. *Every hop strips a qualifier.*

5 What was my prior — and did it move?

Write it down *before* you ask. No prior on paper → no way to know whether you actually updated.

MOVE 1 A prompt is a program.

*A prompt isn't a question — it's a **program**. You're programming a prediction engine with words.*

JULES WHITE, VANDERBILT

You already know this discipline. An **OPORD** is five paragraphs; an **SBAR** is four. A good prompt can be formulaic.

MOVE 2 CO-STAR-E — a scaffold for the prompt.

A starting structure for the ask itself:

C Context

The background it needs. Who's asking, the setting, what's already true.

O Objective

The actual ask — the verb. What you want it to *do*, specifically.

S Style

How it should read. Structure, length, formatting, vocabulary.

T Tone / Persona

The register. Clinical or warm. Hedged or committed.

A Audience

Who's reading. Each reader pulls a different vocabulary.

R Response

The format back. Paragraph, table, list — a structured object you can use.

E Examples

Show, don't tell. One or two beats a paragraph of rules.

CO-STAR from Sheila Teo — GovTech Singapore's first GPT-4 prompt-engineering competition (2023). +E for Examples.

MOVE 3 The higher-leverage move: ground it in context.

CO-STAR-E gets you a clean prompt. The bigger lever is **context** — what you ground the model in. To audit a claim, give it two things: **the claim or paper** you're checking, and **the**

standard you want it judged against — a methods text, run as an adversarial reviewer rather than a cheerleader.

Thinking Clearly with Data

The logic of evidence, end to end — what a comparison can and can't show. Bueno de Mesquita & Fowler.

OpenIntro Statistics (free)

The shared foundation — sampling, inference, what a p-value does and doesn't say. Or any stats text you trust.

Your favorite text — or a key methods paper — goes here.

Bring the standard you already rely on. The method is the same; the rubric is yours.

Works best when the standard is a **PDF**. Hand any PDF you own (or find open-source) to an AI of your choice and ask it to turn it into a **tight, chapter-by-chapter markdown extract**. That's what I do — then I point the audit at the markdown.

Then run a prompt like this — CO-STAR-E doing the work:

ADVERSARIAL AUDIT PROMPT

CONTEXT: I'm auditing a claim, not defending it. Attached: [the paper/claim] and [book.md], a methods text I trust as the standard.

OBJECTIVE: Using [book.md] as the rubric, list every assumption the work makes — explicit and implicit — and check each against the book. Then separate what was measured from what was claimed.

STYLE: One finding per line. Cite the book chapter for each check. No praise for the writing.

TONE: Skeptical and even-handed. Argue both sides. Default to "the evidence doesn't support that" when it doesn't.

AUDIENCE: A clinician who knows the subject but not the statistics.

RESPONSE: A numbered list. Tag each finding [the source's own data] / [inference] / [needs outside evidence].

Stay skeptical, not cynical. The job was never to reject the tool — it's to keep asking the questions.

Dr. Jared Black — Army-Baylor / Graduate School Research & Education Symposium.
From the keynote *Simplicity on the other side of complexity*.